CODSOFT ARTIFICIAL INTELLIGENCE INTERNSHIP

Image Captioning System

- **Submitted By:**
  Neerja Mehar

- **Internship Domain:**
  Artificial Intelligence

- **Organization:**
  CODSOFT

Project: Image Captioning System

<u>Introduction</u>

Image Captioning is the process of generating a textual description of an image using Artificial Intelligence techniques. It combines Computer vision for image understanding and natural Language Processing (NLP) for text generation.

Objectives

- To develop an AI-based Image Captioning System.
- To automatically generates captions from images.
- To apply Computer Vision and Deep Learning techniques.
- To use the BLIP pre-trained model for image understanding.
- To gain practical experience with AI and Machine Learning techniques.

Features

- Automatic image caption generation using AI.
- User-friendly command-line interface.
- Generates meaningful and human readable caption.
- Easy to implement and use.
- Supports image input through file path.
- Supports common image formats such as JPG, JPEG, and PNG.

System Requirements

- **Hardware Requirements:**

  - Processor: Intel i3 or above
  - RAM: 4 GB or higher
  - Storage: 2 GB free space
  - Internet Connection

- **Software Requirements:**

  - Windows 10/11
  - Python 3.x
  - VS code
  - Transformers Library
  - PyTorch
  - Pillow

Tools and Technology Used

- Python
- Hugging Face Transformers
- PyTorch
- Pillow (PIL)
- BLIP pre-trained model
- Visual Studio Code

Algorithm

1. Start the program.
2. Load the BLIP pre-trained model.
3. Accept image path from the user.
4. Check whether the image file exists.
5. Open and process the image.
6. Extract image features using the processor.
7. Generate caption using the model,
8. Display the generated caption.
9. Repeat or exit the program.
10. End

Source Code

```
from transformers import  BlipProcessor,
BlipForConditionalGeneration
from PIL import Image
import torch
import os

print("="* 50)
print("    IMAGE CAPTIONING SYSTEM")
print("=" * 50)

# Load Pretrained Model
print("Loading AI Model... Please Wait...")
```

```
processor =
BlipProcessor.from_pretrained("Salesforce/blip-
image-captioning-base")
model =
BlipForConditionalGeneration.from_pretrained(
    "Salesforce/blip-image-captioning-base"
)

print("Model Loaded Successfully!\n")

def generate_caption(image_path):
    try:
        # if file exists
        if not os.path.exists(image_path):
            print("Error: Image File not found!")
            return

        # open image
        image = Image.open(image_path)

        # Generator Caption
        inputs = processor(image, return_tensors="pt")

        output = model.generate(**inputs)

        caption = processor.decode(output[0],
skip_special_tokens=True)

        print("\nGenerated Caption:")
```

```
        print("-" * 30)
        print(caption)
        print("-" *30)

    except Exception as e:
        print("An Error Occured:", e)

def main():
    while True:
        print("\nOptions:")
        print("1. Generate Caption")
        print("2. Exit")

        choice = input ("Enter your Choice:")

        if choice == "1":
            image_path = input("Enter Image Path:")
            generate_caption(image_path)

        elif choice ==  "2":
            print("Thank you for Usinf Image Captioning 
system!")
            break

        else:
            print ("Invalid Choice! Try Again.")

if __name__=="__main__":
    main()
```

Sample Output

```
Options:
1. Generate Caption
2. Exit
Enter your Choice:1
Enter Image
Path:Task4_ImageCaptioning\Airplane.jpg

Generated Caption:
------------------------------
a large passenger jet sitting on top of an airport tar
------------------------------

Options:
1. Generate Caption
2. Exit
Enter your Choice:2
Thank you for Usinf Image Captioning system!
```

Learning Outcomes

- Learned the fundamentals of image captioning.
- Gained experience with Hugging Face Transformers.
- Understood the working of pre-trained AI models.

- Improved knowledge of Computer Vision and NLP.
- Enhance python programming and debugging skills.

Future Enhancements

- Support for multiple languages in caption generation.
- Development of a Graphical User Interface (GUI).
- Real-time image captioning using webcam input.
- Batch processing for multiple images at once.
- voice output for generated caption.

Conclusion

The Image Captioning System successfully generates meaningful captions for image using Artificial Intelligence. The project demonstrates the practical implementation of Computer vision and Natural Language Processing techniques. It provides an efficient way to describe image content automatically and serves as a useful learning experience in AI and Deep Learning technologies.

Result

The Image Captioning System was successfully

Developed and tested using the BLIP (Bootstrapping Language-Image pre-training) model from Hugging Face.

The system accepts image an automatically generates a meaningful text caption describing the image. During, testing, the model correctly identified the contents of sample images and produced relevant captions.